

SIMATS ENGINEERING ASSESSMENT TOOL3

COURSE CODE: MLA 02 COURSE NAME: Fundamentals Of Machine Learning

COURSE OUTCOME COVERED

CO2: *Analyze unsupervised learning methods, including clustering and dimensionality reduction, to extract insights from unlabelled data.(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 50

Total Marks:50

Annexure A

MCQ Test

Code of Conduct:

I B.Harsha Vardhan(192425201)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:B.Harsha Vardhan



CO3 AT3 MCQ TEST

Total points **49/50** ?

MLA02 MACHINE LEARNING

0 of 0 points

Registration Number: *

192425201

Name: *

B Harsha Vardhan

Answer ALL
Questions

49 of 50
points

Principal components in PCA are: 1/1



Annexure A

Q1. Which clustering algorithm requires the number of clusters to be specified before training?

- A) K-Means
 - B) Gaussian Mixture Model
 - C) DBSCAN
 - D) Hierarchical Clustering
-

Q2. What is the primary objective of the K-Means algorithm?

- A) Maximize inter-cluster variance
 - B) Minimize the sum of squared distances between data points and cluster centroids
 - C) Classify data using labels
 - D) Maximize the number of clusters
-

Q3. Which method is commonly used to determine the optimal number of clusters in K-Means?

- A) Gradient Descent
 - B) Elbow Method
 - C) Cross Validation
 - D) Grid Search
-

Q4. In K-Means clustering, a centroid represents:

- A) The farthest data point in a cluster
 - B) The average position of all data points in a cluster
 - C) A randomly selected feature
 - D) The largest data value
-

Q5. Which distance measure is most commonly used in the standard K-Means algorithm?

- A) Manhattan Distance
 - B) Euclidean Distance
 - C) Cosine Similarity
 - D) Hamming Distance
-

Q6. Which algorithm performs soft clustering by assigning probabilities to cluster memberships?

- A) K-Means
 - B) Gaussian Mixture Model (GMM)
 - C) DBSCAN
 - D) K-Nearest Neighbors
-

Q7. The Expectation-Maximization (EM) algorithm consists of which two main steps?

- A) Training and Testing
 - B) Selection and Mutation
 - C) Expectation and Maximization
 - D) Encoding and Decoding
-

Q8. During the Expectation (E) step, the EM algorithm primarily:

- A) Updates model parameters
 - B) Calculates cluster membership probabilities
 - C) Removes noisy features
 - D) Splits the dataset
-

Q9. During the Maximization (M) step, the EM algorithm primarily:

- A) Updates the model parameters
 - B) Calculates Euclidean distance
 - C) Reduces dimensions
 - D) Selects important features
-

Q10. Gaussian Mixture Models assume that data is generated from:

- A) A single Gaussian distribution
 - B) Multiple Gaussian distributions
 - C) A decision tree model
 - D) A regression equation
-

Q11. Which clustering technique can model clusters with different shapes and sizes more effectively than K-Means?

- A) K-Means
 - B) Gaussian Mixture Model
 - C) Linear Regression
 - D) Logistic Regression
-

Q12. What does the covariance matrix represent in a Gaussian Mixture Model?

- A) Feature names
 - B) Variability and relationship among variables
 - C) Number of clusters
 - D) Missing values
-

Q13. The Curse of Dimensionality mainly occurs when:

- A) The dataset has too many rows
 - B) The dataset has a large number of features
 - C) The dataset contains categorical data
 - D) The dataset has missing values
-

Q14. One major effect of the Curse of Dimensionality is:

- A) Faster model training
 - B) Increased computational complexity
 - C) Reduced storage requirements
 - D) Improved prediction accuracy
-

Q15. Which technique is commonly used to address the Curse of Dimensionality?

- A) Feature Scaling
 - B) Dimensionality Reduction
 - C) Data Duplication
 - D) Oversampling
-

Q16. Dimensionality reduction helps by:

- A) Increasing the number of features
 - B) Removing redundant information
 - C) Creating duplicate records
 - D) Increasing model complexity
-

Q17. Principal Component Analysis (PCA) is primarily used for:

- A) Classification
 - B) Dimensionality Reduction
 - C) Clustering only
 - D) Regression
-

Q18. PCA transforms the original variables into:

- A) Decision Trees
 - B) Principal Components
 - C) Hidden Layers
 - D) Support Vectors
-

Q19. Principal components in PCA are:

- A) Dependent on each other
 - B) Orthogonal to each other
 - C) Randomly selected
 - D) Binary variables
-

Q20. The first principal component captures:

- A) The minimum variance
 - B) The maximum variance in the dataset
 - C) Only categorical features
 - D) The class labels
-

Q21. Which matrix is used in PCA to compute principal components?

- A) Identity Matrix
 - B) Covariance Matrix
 - C) Sparse Matrix
 - D) Confusion Matrix
-

Q22. Eigenvalues in PCA indicate:

- A) Number of observations
 - B) Amount of variance explained by each component
 - C) Cluster labels
 - D) Feature names
-

Q23. Eigenvectors in PCA represent:

- A) Principal component directions
 - B) Class labels
 - C) Cluster centroids
 - D) Hidden variables
-

Q24. PCA is an example of:

- A) Supervised Learning
- B) Unsupervised Learning

- C) Reinforcement Learning
- D) Semi-supervised Learning

Q25. Which preprocessing step is generally recommended before applying PCA?

- A) Data Standardization
- B) Label Encoding
- C) Feature Duplication
- D) Image Augmentation

Q26. Factor Analysis primarily aims to identify:

- A) Hidden (latent) variables
- B) Decision boundaries
- C) Class labels
- D) Outliers

Q27. Factor Analysis assumes that observed variables are influenced by:

- A) Random forests
- B) Latent factors
- C) Neural networks
- D) Clusters only

Q28. Which technique is mainly used for identifying latent constructs in data?

- A) PCA
- B) Factor Analysis
- C) Linear Regression
- D) KNN

Q29. Independent Component Analysis (ICA) attempts to find:

- A) Correlated components
- B) Statistically independent components
- C) Principal components only
- D) Decision boundaries

Q30. ICA is widely used in:

- A) Image Compression
- B) Blind Source Separation
- C) Linear Regression
- D) Data Sorting

Q31. Which of the following assumes that source signals are statistically independent?

- A) PCA
- B) ICA
- C) K-Means
- D) Logistic Regression

Q32. Probabilistic PCA differs from conventional PCA because it:

- A) Uses a probabilistic model
- B) Requires labeled data

- C) Uses decision trees
 - D) Eliminates covariance
-

Q33. Probabilistic PCA is especially useful when handling:

- A) Missing or noisy data
 - B) Text classification
 - C) Image segmentation only
 - D) Database indexing
-

Q34. Which dimensionality reduction technique maximizes variance?

- A) PCA
 - B) ICA
 - C) Factor Analysis
 - D) K-Means
-

Q35. Which technique emphasizes statistical independence rather than variance?

- A) PCA
 - B) ICA
 - C) GMM
 - D) K-Means
-

Q36. Which clustering algorithm assigns each data point to exactly one cluster?

- A) K-Means
 - B) Gaussian Mixture Model
 - C) Fuzzy C-Means
 - D) EM Algorithm
-

Q37. In Gaussian Mixture Models, cluster membership is represented by:

- A) Binary values
 - B) Probability values
 - C) Decision rules
 - D) Distance only
-

Q38. Which algorithm is iterative and converges when parameter updates become negligible?

- A) EM Algorithm
 - B) Breadth-First Search
 - C) Linear Search
 - D) Merge Sort
-

Q39. Which of the following is NOT a dimensionality reduction technique?

- A) PCA
 - B) ICA
 - C) Factor Analysis
 - D) K-Means
-

Q40. The primary benefit of dimensionality reduction is:

- A) Increased feature redundancy
 - B) Reduced computational cost
 - C) Increased class labels
 - D) Increased memory usage
-

Q41. PCA reduces dimensions while preserving maximum:

- A) Correlation
 - B) Variance
 - C) Distance
 - D) Entropy
-

Q42. Which algorithm updates cluster centroids repeatedly until convergence?

- A) K-Means
 - B) PCA
 - C) ICA
 - D) Decision Tree
-

Q43. Which measure is minimized in K-Means clustering?

- A) Within-cluster Sum of Squares (WCSS)
 - B) Entropy
 - C) Information Gain
 - D) Gini Index
-

Q44. Which technique is most suitable for separating mixed audio signals?

- A) PCA
 - B) ICA
 - C) K-Means
 - D) Naïve Bayes
-

Q45. Which method transforms correlated variables into uncorrelated components?

- A) PCA
 - B) EM Algorithm
 - C) GMM
 - D) Factor Analysis
-

Q46. Which of the following is a probabilistic clustering algorithm?

- A) Gaussian Mixture Model
 - B) K-Means
 - C) Decision Tree
 - D) Random Forest
-

Q47. Which technique is commonly used before visualization of high-dimensional data?

- A) PCA
 - B) Linear Regression
 - C) Logistic Regression
 - D) Naïve Bayes
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Q48. The main objective of Factor Analysis is to:

- A) Reduce variables by identifying latent factors
- B) Predict target values
- C) Build classification models
- D) Sort datasets

Q49. Which algorithm can assign a data point to multiple clusters with different probabilities?

- A) Gaussian Mixture Model
 - B) K-Means
 - C) KNN
 - D) Decision Tree
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Q50. Which of the following is an unsupervised machine learning technique?

- A) K-Means Clustering
- B) Logistic Regression
- C) Decision Tree Classification
- D) Support Vector Machine

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time

Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions
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